

Vijay Rameshkumar

Strategic and innovation-driven Data Science Consultant and Machine Learning Engineer experience in building and deploying AI-driven, predictive, and agentic systems that enhance efficiency and profitability and intelligent automation solutions. Targeting roles in Data Science, Machine Learning Engineering, Generative AI, and Agentic AI Systems Development, contributing through scalable, production-ready AI solutions that create measurable business impact.

CONTACT DETAILS

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CORE COMPETENCIES

- Data Science Consulting & AI Strategy
- Machine Learning Engineering & Model Deployment
- AI and RAG Pipelines
- Generative AI and Agentic AI Systems Development
- Predictive Maintenance & Intelligent Automation
- Recommender Systems & Optimization Algorithms
- Deep Learning, Neural Networks & Transfer Learning
- Big Data Analytics & OCR and Intelligent Document Processing (IDP)
- Natural Language Processing (NLP) & Computer Vision (CV)
- Statistical, Predictive & Time Series Modelling
- End-to-End ML Pipeline Automation
- Stakeholder Engagement & Data-Driven Decision Making

EDUCATION

- **MS Computer Science (AI & ML)**
Woolf / Scaler University
2024 – 2026
- **Post Graduate Program in AI/ML -**
Purdue University (2024)
- **Bachelor of Engineering (B.E.) in Mechatronics**
Anna University, Chennai, IN
2013 – 2017

PROFILE SUMMARY

- Accomplished **Data Science and Machine Learning professional** with over 7 years of experience delivering AI-driven, predictive, and intelligent automation solutions across maritime, telecom, and big data domains.
- Skilled in developing **RAG solutions, Generative and Agentic AI architectures**, integrating multi-model intelligence for autonomous decision-making and real-time analytics.
- Proficient in **Python, TensorFlow, Keras, Azure ML, AWS, and GCP**, with expertise in ML pipeline automation, feature engineering, and stakeholder collaboration to drive strategic insights.
- **Strategic AI & Data Science Leader** driving **\$3.5M+** in cost savings through cross-functional collaboration and innovative ML solutions across **maritime, logistics, and telecom domains**; adept at crafting and presenting **AI roadmaps** to executive stakeholders to align technology with business impact.

TECHNICAL SKILLS

- **Machine Learning & AI:** Deep Learning, Neural Network Architecture, Time Series Forecasting, Anomaly Detection, Predictive Modeling, Statistical Analysis, Explanatory Modeling, NLP, Recommender Systems, MLOps, Model Optimization, A/B Testing
- **Generative AI:** LLMs (GPT, BERT, Transformers), RAG Pipelines, Graph RAG (AST-RAG), LangChain, LangGraph, Agentic Workflows, Prompt Engineering, LoRA Fine-tuning, GANs, VAEs, Embeddings, Vector Databases, Multimodal AI, Agentic AI Frameworks, Document Processing, Information Extraction
- **Computer Vision:** YOLO, R-CNN, Image Segmentation, Object Tracking, Pose Detection
- **NLP:** BERT, GPT, Transformers, Sentiment Analysis, Text Classification
- **Time Series & Recommender Systems:** LSTM, ARIMA, Prophet, Anomaly Detection, Supplier Recommender Systems
- **Data Visualization & Business Intelligence:** Power BI, Interactive Dashboards, Real-time Dashboards, Matplotlib, Seaborn, Bokeh, Plotly, Python APIs, Enterprise Reporting
- **Programming Languages:** Python (TensorFlow 2.x, PyTorch, Keras, Scikit-learn, Pandas, NumPy, Statsmodels), SQL, C++
- **Cloud Platforms:** AWS (SageMaker, Lambda, EC2, S3, ECS, ECR, API Gateway, DynamoDB, CloudWatch), Azure (ML Studio, OpenAI Services, Databricks, Cognitive Services)
- **Data Management:** PostgreSQL, MySQL, MongoDB, NoSQL, Snowflake, Apache Spark, Databricks, Vespa Vector DB
- **MLOps & Infrastructure:** MLFlow, Airflow, Docker, ECR, ECS, Model Registry, CI/CD Pipelines, Grafana, Elasticsearch

WORK EXPERIENCE

Miracle Software Private Limited (FORD), Chennai | Senior AI/ML Engineer Apr'26 – Present

- Analyzed complex program data to uncover business improvement opportunities and support strategic decision-making.
- Designed and implemented a vehicle program tracking platform covering design, engineering, manufacturing, launch, marketing, and warranty phases.
- Applied machine learning models to forecast warranty risks and detect schedule bottlenecks before critical milestones.
- Developed early-warning and alerting mechanisms for program managers, enabling timely intervention and reducing project delays.
- Improved cross-functional visibility and execution efficiency through data-driven monitoring and predictive analytics.

PRODUCT CONTRIBUTIONS

- **Propulsion Pro AI:** Real-time conversational AI for vessel troubleshooting and root-cause analysis across 500+ ships.
- **Ocean Eye Agentic AI Platform:** Automated maritime workflows, cutting manual document handling by 80%.
- **Enterprise RAG Chatbot:** AI chatbot for voyage tracking and performance insights with document intelligence.
- **Supply Chain Analytics Dashboard:** Optimized supply chain efficiency, reducing bottlenecks by 25%.

ACHIEVEMENTS

- **Predictive Maintenance:** Cost savings of **USD 1.2 million/year** through optimized asset efficiency.
- **Vendor Recommender System:** Achieved **USD 3 million/month** procurement cost savings across **564 vessels**.
- Developed a **predictive maintenance model** that cut equipment downtime by 10%, boosting overall efficiency.
- Implemented **data visualization techniques** improving report clarity and **increasing stakeholder engagement**.
- Built a **data pipeline** that streamlined ship performance processing, reducing turnaround time by 30%.
- Enhanced **model accuracy** with advanced feature engineering, improving predictive performance 50%.

CERTIFICATIONS

- **Post Graduate Program in AI/ML -** Purdue University (2024)
- **Practical Time Series Analysis – State** University of New York (2021)
- **Applied AI Certified – Advanced Machine Learning (Professional Development) (2021)**
- **Edureka – IoT Solution Engineer Certification (2019)**

PROFESSIONAL COURSES

- Data Science Master's Program – Applied AI (Advanced AI)
- Hope Tutors – AI (Machine Learning) Certification

Exavalu Private Limited, Chennai | Tech Lead - AI Oct'25 – Jan'26

- Leading development of Intelligent Document Processing (IDP) solutions for insurance automation.
- Managing a 10-member product team, ensuring efficient delivery and innovation.

Synergy Maritime, Chennai | Senior Data Scientist Jan'23 – Oct'25

- Analyzed extensive ship performance datasets to derive actionable insights and **recommended optimal vessel speeds**, enhancing engine efficiency and minimizing fuel consumption and emissions.
- Developed and deployed a **Hybrid Vessel Scorecard in Azure ML**, integrating real-time performance metrics for fleet-wide operational benchmarking and Leading **Intelligent Document Processing (IDP)** solutions for insurance automation using **OCR, NLP, and LLM** technologies.
- Collaborated with cross-functional marine operations teams to **translate analytical outputs** into data-backed strategies for improving vessel sustainability & cost efficiency.
- Engineered and deployed AI-driven predictive maintenance models that enhanced asset efficiency and achieved **annual savings of USD 1.2 million** through reduced downtime and optimized vessel operations.
- **Delivered measurable business value** with **USD 3 million monthly cost savings** across **564 vessels**, improving sourcing efficiency and contract utilization.
- Designed **forecasting models** to support real-time operational analytics and improve decision-making accuracy across maritime performance management systems.
- Integrated multi-model **Agentic AI frameworks** & product prototypes, enabling autonomous analytics & intelligent process automation as an **independent developer**.

Value Labs, Hyderabad (WFH) | Data Scientist Apr'22 – Jan'23

- Investigated real-time supply chain data to identify key patterns and **analyze event occurrences** impacting procurement and logistics performance.
- Designed and implemented a **Hybrid Recommender System** leveraging machine learning techniques to enhance vendor selection and supply chain optimization.
- Deployed the recommendation model on **AWS**, ensuring **scalability, reliability, and seamless integration** with enterprise applications for business decision-making.

CanGo Networks Telecommunications, Chennai | Data Scientist Apr'21 – Mar'22

- Executed comprehensive research on real-time telecom fault prediction and event occurrence, applying advanced ML techniques to **engineer dynamic predictive models** and **reduce network downtime**.
- Devised and implemented an AR-based multi-KPI optimization algorithm inspired by published studies, enabling unified monitoring and **performance enhancement**.
- Engineered a graph-based fault prediction framework to **anticipate device failures** across large-scale telecom networks, improving reliability & preventive maintenance.
- Developed an intelligent recommendation engine for telecom operators by conducting **root cause analysis (RCA)** and suggesting data-driven corrective actions.
- Delivered impactful insights and model outcomes to clients and stakeholders, **influencing strategic decision-making** and optimizing operational efficiency.

Machine Origin, Bangalore | Data Scientist Nov'19 – Nov'20

- Translated **business requirements into ML problem statements**, designing end-to-end computer vision models from concept to deployment.
- Developed and validated **deep learning models** for detecting abnormal activities in live CCTV footage using **classification, detection, segmentation, tracking, and pose estimation** algorithms.

- Conducted **experimentation with multiple modeling approaches** to optimize accuracy and real-time performance across surveillance datasets.
- Collaborated with cross-functional teams to **integrate computer vision pipelines** into production environments, enhancing situational awareness & automation efficiency.

Suriyan Innovative LLP, Bangalore | Junior Robotics / Automation Engineer Jun'18 – Jun'18

- Designed & developed **IoT-enabled robotic systems** by integrating embedded hardware, 3D printing, & cloud connectivity to support industrial automation applications.
- Built **end-to-end IoT solutions** using **ESP32, Arduino, & Raspberry Pi** microcontrollers with **MQTT protocol**, enabling seamless device-to-cloud communication & remote monitoring.
- **Integrated 50+ sensor types** (temperature, pressure, proximity, accelerometer, gyroscope) through **datasheet-driven configuration**, ensuring accurate data acquisition and processing.
- Developed **precision motor control systems** utilizing **stepper and servo motors, H-bridge circuits, & PWM control** to optimize robotic movement & automation performance.
- Engineered **secure IoT device provisioning workflows** using **SSH key-based authentication** and **encrypted data transmission**, enhancing system reliability and data protection.
- Created real-time monitoring dashboards via **Python APIs**, integrating IoT devices with **cloud platforms (BLE and Wi-Fi connectivity)** for live visualization, control, and analytics.

Annexmed Pvt. Ltd., Chennai | Claim Processing Executive May'17 – Nov'17

- Processed and validated **medical insurance claims and appeals** for US-based clients, ensuring accuracy and compliance.
- Analyzed claim discrepancies to **reduce rejections and accelerate reimbursements**, improving turnaround efficiency.
- Coordinated with internal teams to **streamline claim workflows** and enhance client satisfaction through timely resolutions.